

4-Way Intersection Control — Raspberry Pi 5

Assignment 3 Short Report

Team Name: _____ Student IDs: _____

1. Design

The controller treats North/South as one non-conflicting traffic group and East/West as the other. The program always starts in an all-red state. It then follows the sequence START/ALL RED → NORTH-SOUTH GREEN → NORTH-SOUTH YELLOW → ALL RED/CLEARANCE → EAST-WEST GREEN → EAST-WEST YELLOW → ALL RED/CLEARANCE → repeat. The code uses functions, a LIGHTS dictionary, a continuous loop, safety checking, and try/except/finally handling.

2. Timing Choices

State	Duration	LED condition
Startup ALL RED	2 s	All approaches red
NS GREEN	10 s	N/S green; E/W red
NS YELLOW	2 s	N/S yellow; E/W red
ALL RED clearance	1 s	All approaches red
EW GREEN	10 s	E/W green; N/S red
EW YELLOW	2 s	E/W yellow; N/S red
ALL RED clearance	1 s	All approaches red

The assignment states that the 10-second requirement applies to the GREEN right-of-way phase, so GREEN_TIME is set to 10 seconds. Yellow and the 1-second all-red clearance are additional transition time. The all-red interval was selected as a brief safety clearance before the conflicting group receives green.

3. Edge Cases and Safety

Edge case	Design response
Power-on/startup	Initialize the system in ALL RED and hold for 2 seconds before normal operation.
Conflicting greens	The state sequence never requests both groups green. verify_no_conflicting_greens() checks the green outputs and forces A
Green → Yellow → Red	Every green phase is followed by its yellow phase; the next conflicting green is only enabled after ALL RED clearance.
All-red clearance	A 1-second ALL RED interval is used between groups to provide an explicit safety buffer.
Ctrl+C during green	KeyboardInterrupt is caught and finally calls safe_shutdown(), which attempts to put every direction RED.
Unexpected exception	The general Exception handler reports the error; finally still attempts safe shutdown.
Wrong pin / loose wire	Run individual LED tests first, compare each LED against the BCM pin table, check resistor/wire continuity, and verify ground
LED failure	The software cannot reliably detect a physically failed LED in this simple circuit. A real safety-critical controller would need re
Timing/blocking	The program uses only the intended sleep() intervals in the state sequence. No unnecessary delay is placed inside the state-
Restart after failure	The program executes all_red() at startup, so every restart begins from a known safe software state.

4. Testing Plan and Results

The following tests must be performed on the physical Raspberry Pi and breadboard. The results below are intentionally left blank because they require real hardware observations.

Test	Expected result	Observed result / evidence
Individual LEDs	Each of 12 LEDs turns on with the correct BCM pin.	_____
All four red LEDs	N/S/E/W red simultaneously.	_____
North-South group	N/S green together; E/W red.	_____
East-West group	E/W green together; N/S red.	_____
Three full cycles	Sequence repeats correctly with no conflicting greens.	_____
Ctrl+C during green	Program leaves intersection ALL RED.	_____
Restart	Program begins ALL RED before first green.	_____
Edge-case test #5+	Document additional fault/wiring/timing test.	_____

5. Troubleshooting Notes

If an LED does not respond, stop the program and check the BCM number, LED orientation, resistor connection, jumper wire, and common ground. Test one LED at a time before running the complete controller. Do not connect an LED directly to a Raspberry Pi GPIO without its current-limiting resistor.

6. Evidence to Attach

Attach the circuit diagram, state diagram, photos/screenshots of testing, a photo of the working Raspberry Pi prototype, and a clear video showing the prototype operating. The final submission also requires an in-class demonstration of at least two complete cycles.

7. Submission Checklist

- traffic_intersection.py
- circuit_diagram.png
- state_diagram.png
- report.pdf completed with actual test results
- prototype photo
- Video/prototype.mp4
- Team name and student IDs added to Python file/report
- Demonstrate at least two complete cycles in class
- Zip folder as GroupName_StudentIDs.zip
- Complete teacher check-off within the required one-week period